

Functions and Function Notation



Function notation

- 1 Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0)$	b $f(2)$	c $f(-1)$	d $f(\frac{1}{2})$
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 - 2 Consider the function $g(x) = \sqrt{x + 5}$. Evaluate:

a $g(4)$	b $g(-1)$	c $g(11)$	d $g(-5)$
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 - 3 Consider the function $h(x) = \frac{x-3}{x+2}$. Evaluate:

a $h(0)$	b $h(3)$	c $h(-1)$	d $h(5)$
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 - 4 If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a .
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Identifying functions

- 5 Determine whether each equation defines y as a function of x :

a $y = 2x + 1$	b $x = y^2$
c $y = x $	d $x^2 + y^2 = 9$
 - 6 Explain how the vertical line test is used to decide whether a graph represents a function.
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Combining functions

- 7 Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f + g)(x)$	b $(f - g)(x)$
c $(f \cdot g)(2)$	d $(f \circ g)(1)$
- 8 A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

a Interpret the value $C(0)$ in context.	b Find $C(12)$.
c Solve $C(n) = 925$ and interpret your answer.	